

Sobolev Tensors Always Belong to the Tensor Product of Their Minimal Subspaces

Candidate full resolution for arXiv:1809.11001

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Status: candidate full solution, likely valid pending human review. The conclusion of Proposition 3.7 of the source holds without Assumption 3.5; therefore the counterexamples conjectured in Remark 3.6 do not exist.

1 The source conjecture and the result

Let $\Omega = \prod_{j=1}^d \Omega_j$. Ali and Nouy identify

$$H^1(\Omega) = \bigcap_{k=1}^d \left(H^1(\Omega_k) \widehat{\otimes}_2 \widehat{\bigotimes_{j \neq k} L^2(\Omega_j)} \right), \quad (1)$$

with an equivalent intersection norm. For $u \in H^1(\Omega)$, they define $U_j^{\min}(u)$ as the $H^1(\Omega_j)$ -closure of all contractions of u against L^2 functionals in the other variables. Proposition 3.7 proves

$$u \in \overline{\bigotimes_{j=1}^d {}^a U_j^{\min}(u)}^{H^1(\Omega)} \quad (2)$$

under Assumption 3.5: the H^1 -orthogonal projection onto every $U_j^{\min}(u)$ must be bounded in L^2 . Remark 3.6 conjectures that (2) can fail when this assumption fails [1].

With the preceding lemma we may now define $U_j^{\min}(u) = \overline{U_a^j(u)}^{\|\cdot\|_1} \subset H^1(\Omega_j)$. This space differs from $U^j(u)$ in case $u \notin {}_a\bigotimes_{j=1}^d H^1(\Omega_j)$. We want to check if property (2.2) still holds for H^1 functions. To this end, we require the following assumption.

Assumption 3.5. *Let $\mathcal{P}^j : H^1(\Omega_j) \rightarrow U_j^{\min}(u)$ be an orthogonal projection. We assume $\mathcal{P}^j$ is continuous in L^2*

$$\sup_{\substack{v_j \in H^1(\Omega_j), \\ v \neq 0}} \frac{\|\mathcal{P}^j v_j\|_0}{\|v\|_0} < \infty.$$

Remark 3.6. *We will frequently encounter Assumption 3.5 in the following sections. We will discuss sufficient conditions for this assumption to be satisfied in part II of this series. We will see that this assumption is not necessarily satisfied. In fact, we conjecture that there are functions $u \in H^1(\Omega)$ which do not satisfy the statement of Theorem 3.7. The proof of this, however, seems to be not trivial.*

Proposition 3.7. *Let $u \in H^1(\Omega)$ and assume 3.5 is satisfied. Then, it holds $u \in \|\cdot\|_1 \bigotimes_{j=1}^d U_j^{\min}(u)$.*

Proof. Since $u \in H^1(\Omega)$, by (2.1), $u \in H^1(\Omega_j) \otimes_{\|\cdot\|_{e_j}} \left({}_a\bigotimes_{k \neq j} L^2(\Omega_k) \right)$. The space $U_j^{\min}(u) \subset H^1(\Omega_j)$ is a closed subspace of the Hilbert space $H^1(\Omega_j)$. Thus, $U_j^{\min}(u) \in \mathbb{G}(H^1(\Omega_j))$. Moreover, $\|\cdot\|_{e_j}$ is a uniform crossnorm. This holds for any $1 \leq j \leq d$ and thus by [5] Theorem 6.29] we obtain

$$u \in \bigcap_{k=1}^d U_k^{\min}(u) \otimes_{\|\cdot\|_{e_k}} \left({}_a\bigotimes_{l \neq k} L^2(\Omega_l) \right).$$

Figure 1: The definition, Assumption 3.5, conjecture, and conditional proposition in arXiv:1809.11001, PDF page 9.

The conjecture is false: (2) always holds. The source projection need not extend to L^2 . Instead, one spectrally truncates the inclusion of the minimal subspace into its L^2 closure. The truncated projections are simultaneously contractive in both norms, and their ranges have equivalent H^1 and L^2 norms. This makes finite tensors dense after truncation; letting the spectral threshold tend to zero proves (2).

We first prove the mechanism for arbitrary Hilbert pairs. This also makes clear that compactness of the Sobolev embedding is unnecessary.

2 A simultaneous spectral-cutoff lemma

Lemma 1 (two-norm cutoffs). *Let U and M be Hilbert spaces and let $J : U \rightarrow M$ be continuous, injective, and have dense range. There are closed subspaces $S_\varepsilon \subset U$ and operators $E_\varepsilon : U \rightarrow U$, $\tilde{E}_\varepsilon : M \rightarrow M$ for $\varepsilon > 0$ such that:*

1. E_ε and $\tilde{E}_\varepsilon$ are orthogonal projections in U and M , respectively, and $\tilde{E}_\varepsilon J = J E_\varepsilon$;
2. both projections are contractions in their own Hilbert norms;
3. on $S_\varepsilon = \text{Ran } E_\varepsilon$,

$$\sqrt{\varepsilon} \|x\|_U \leq \|Jx\|_M \leq \|J\| \|x\|_U; \tag{3}$$

4. $E_\varepsilon \rightarrow I_U$ strongly in U , and $\tilde{E}_\varepsilon \rightarrow I_M$ strongly in M , as $\varepsilon \downarrow 0$.

Proof. Put $K = J^* J$. This is a bounded positive self-adjoint operator on U and $\ker K = \ker J = \{0\}$. By the spectral theorem, set

$$E_\varepsilon = \mathbf{1}_{[\varepsilon, \infty)}(K), \quad S_\varepsilon = \text{Ran } E_\varepsilon. \tag{4}$$

It is an orthogonal projection in U , and for $x \in U$,

$$\|JE_\varepsilon x\|_M^2 = \langle KE_\varepsilon x, E_\varepsilon x \rangle_U \leq \langle Kx, x \rangle_U = \|Jx\|_M^2. \quad (5)$$

Furthermore, if $x \in S_\varepsilon$, spectral localization gives $\|Jx\|_M^2 \geq \varepsilon \|x\|_U^2$, proving (3).

The range $J(S_\varepsilon)$ is closed in M : an M -Cauchy sequence in that range is U -Cauchy by (3), and its U -limit remains in the closed subspace S_ε . Since E_ε commutes with K ,

$$\langle JE_\varepsilon x, Jy \rangle_M = \langle KE_\varepsilon x, y \rangle_U = \langle Jx, JE_\varepsilon y \rangle_M. \quad (6)$$

Thus (5) lets $JE_\varepsilon J^{-1}$ extend from the dense subspace $J(U)$ to the M -orthogonal projection $\tilde{E}_\varepsilon$ onto $J(S_\varepsilon)$.

Because K has no spectral mass at zero, $E_\varepsilon x \rightarrow x$ in U for every $x \in U$. The continuous inclusion then gives convergence in M on the dense set $J(U)$; the contraction bound and density extend it to every element of M . $\square$

3 The abstract intersection theorem

Let $A_j \hookrightarrow H_j$ be continuous injective dense embeddings of Hilbert spaces, $1 \leq j \leq d$. All tensor products below are canonical Hilbert tensor products. Inside $\mathcal{H} = \bigotimes_{j=1}^d H_j$, set

$$X_k = A_k \widehat{\otimes}_2 \widehat{\bigotimes_{j \neq k} H_j}, \quad \mathcal{X} = \bigcap_{k=1}^d X_k, \quad \|v\|_{\mathcal{X}} = \max_k \|v\|_{X_k}. \quad (7)$$

For $u \in \mathcal{X}$, let $H_{[j]} = \widehat{\bigotimes_{i \neq j} H_i}$ and define

$$U_j = \overline{\{(I_{A_j} \otimes \ell)u : \ell \in H_{[j]}^*\}}^{A_j}. \quad (8)$$

It is equivalent in (8) to use only finite linear combinations of decomposable functionals, since these are dense in $H_{[j]}^*$ and the contraction map $H_{[j]}^* \rightarrow A_j$ is continuous.

Theorem 2 (minimal subspaces in a Hilbert intersection). *In the setting above, every $u \in \mathcal{X}$ satisfies*

$$u \in \overline{\bigotimes_{j=1}^d U_j}^{\mathcal{X}}. \quad (9)$$

Proof. We split the proof into three steps.

Step 1: put all passive factors into their H -closures. Let

$$M_j = \overline{U_j}^{H_j}. \quad (10)$$

Viewing u in $X_k = A_k \widehat{\otimes}_2 H_{[k]}$, the usual Hilbert-tensor minimality argument gives

$$u \in U_k \widehat{\otimes}_2 H_{[k]}. \quad (11)$$

For completeness, if P_k is the A_k -orthogonal projection onto U_k , then every contraction of $((I - P_k) \otimes I)u$ against an element of $H_{[k]}^*$ vanishes by (8); hence the tensor itself vanishes, proving (11).

Let Q_j be the H_j -orthogonal projection onto M_j . The same contraction argument in $\mathcal{H}$ shows $(Q_j \otimes I)u = u$ for every j , so

$$u \in \widehat{\bigotimes_{j=1}^d M_j}. \quad (12)$$

Fix k . For $j \neq k$, Q_j acts contractively on the corresponding H_j factor of $U_k \widehat{\otimes}_2 H_{[k]}$. Applying their tensor product to (11), and using (12), gives

$$u \in Y_k := U_k \widehat{\otimes}_2 \widehat{\bigotimes_{j \neq k} M_j} \quad (1 \leq k \leq d). \quad (13)$$

Equality first holds in the common ambient space $\mathcal{H}$ and therefore in X_k , whose embedding into $\mathcal{H}$ is injective.

Step 2: construct one cutoff that converges in every directional norm. Apply Lemma 1 to the inclusion $J_j : U_j \hookrightarrow M_j$. Denote the resulting projections by $E_{j,\varepsilon}$ on U_j and $\tilde{E}_{j,\varepsilon}$ on M_j , and their common range by $S_{j,\varepsilon} \subset U_j$. On Y_k define

$$T_{k,\varepsilon} = E_{k,\varepsilon} \widehat{\otimes}_2 \widehat{\bigotimes_{j \neq k} \tilde{E}_{j,\varepsilon}}. \quad (14)$$

It is a contraction: the active factor is contractive in A_k , and every passive factor is contractive in H_j . Strong convergence of finite Hilbert tensor products and (13) yield

$$T_{k,\varepsilon} u \longrightarrow u \quad \text{in } X_k. \quad (15)$$

The tensors in (14) are all restrictions of the same H -orthogonal product projection $\widehat{\bigotimes_j \tilde{E}_{j,\varepsilon}}$ on $\widehat{\bigotimes_j M_j}$. Consequently their values at u agree in $\mathcal{H}$; call this common value u_ε . Since there are finitely many k , (15) gives

$$\|u_\varepsilon - u\|_{\mathcal{X}} \longrightarrow 0. \quad (16)$$

Step 3: approximate each cutoff by finite tensors. For fixed ε , Lemma 1 says that the A_j and H_j norms are equivalent on $S_{j,\varepsilon}$. Hence the d Hilbert tensor norms

$$S_{k,\varepsilon}^{A_k} \widehat{\otimes}_2 \widehat{\bigotimes_{j \neq k} S_{j,\varepsilon}^{H_j}}, \quad 1 \leq k \leq d, \quad (17)$$

are mutually equivalent on the algebraic tensor product $\bigotimes_j^a S_{j,\varepsilon}$. Their completions therefore coincide, with equivalent norms. By (14), u_ε belongs to that common completion. It can thus be approximated simultaneously in every X_k norm by finite tensors from

$$\bigotimes_{j=1}^d {}^a S_{j,\varepsilon} \subseteq \bigotimes_{j=1}^d {}^a U_j. \quad (18)$$

Choose $\varepsilon_n \downarrow 0$ and, using (18), choose v_n with $\|v_n - u_{\varepsilon_n}\|_{\mathcal{X}} < 1/n$. Equation (16) then gives $v_n \rightarrow u$ in $\mathcal{X}$, which is (9). $\square$

4 Resolution of the Sobolev conjecture

Corollary 3 (unconditional form of Proposition 3.7). *In the setting of arXiv:1809.11001, let $\Omega = \prod_{j=1}^d \Omega_j$ and $u \in H^1(\Omega)$. Then*

$$u \in \overline{\bigotimes_{j=1}^d {}^a U_j^{\min}(u)}}^{H^1(\Omega)}. \quad (19)$$

No L^2 -boundedness assumption on the H^1 -orthogonal projections is required.

Proof. Take $A_j = H^1(\Omega_j)$ and $H_j = L^2(\Omega_j)$. The source’s identification (1) is precisely (7), up to an equivalent intersection norm. The source defines $U_j^{\min}(u)$ using decomposable L^2 functionals in the other factors. As observed after (8), completing those functionals gives the same A_j -closed subspace. Theorem 2 now gives (19). $\square$

Thus the membership statement conjectured to fail in Remark 3.6 is in fact universal. This does not say that Assumption 3.5 itself is automatic: the full H^1 -orthogonal projection onto $U_j^{\min}(u)$ can still be unbounded in L^2 . The proof only constructs an increasing family of two-norm-bounded spectral cutoffs, which is enough for approximation of the single tensor u .

5 Checks, scope, and novelty

The proof does not use compactness of $H^1(\Omega_j) \hookrightarrow L^2(\Omega_j)$, eigenfunction expansions, or a uniform bound as $\varepsilon \downarrow 0$. Continuous spectrum causes no problem: strong convergence in Lemma 1 follows from $\ker(J_j^* J_j) = 0$. The only structural inputs are the finite intersection (1), Hilbert tensor products, and continuous injective component embeddings.

Part II repeats that membership was known only under conditions and says the authors did not believe it held generally [2]. The earlier work of Falcó and Hackbusch proves density of the algebraic product of the full component Sobolev spaces in the Sobolev intersection [3]. That density does not put the factors into the tensor-dependent minimal subspaces; the spectral cutoff in this packet supplies exactly that missing localization. A bounded search through 13 August 2026 found the two source papers and the general minimal-subspace literature, but no later paper giving (19) or the abstract cutoff theorem above. This is bounded evidence, not an exhaustive novelty claim.

Human review should focus on three points: (i) the refinement (13) in the common L^2 ambient tensor product; (ii) the M -orthogonal extension in Lemma 1; and (iii) the identification of the completions in (17). Each is proved explicitly above.

References

- [1] M. Ali and A. Nouy, *Singular Value Decomposition in Sobolev Spaces: Part I*, arXiv:1809.11001v3 (2020).
- [2] M. Ali and A. Nouy, *Singular Value Decomposition in Sobolev Spaces: Part II*, arXiv:1912.11293 (2019).
- [3] A. Falcó and W. Hackbusch, *On minimal subspaces in tensor representations*, Foundations of Computational Mathematics **12** (2012), 765–803.